



© Institut für Energietechnik und Thermodynamik, Technische Universität Wien

Numerical investigation of added mass and damping effects on a hydrofoil in cavitation tunnel

The added effects such as added mass and damping acting from the fluid on a structure, strongly affect the dynamic response of the structure submerged in water. The main goal of this paper is to present preliminary numerical investigations of added mass and damping effects acting on a hydrofoil placed in the cavitation tunnel test section which has been designed and built at Viktor Kaplan Department of Fluid Engineering at Brno University of Technology. In this paper, the natural frequencies shift and **linear dependence of the damping ratio on the fluid velocity have been found.**

Pavel Čupr, Pavel Rudolf and Vladimír Habán

1 Introduction

The phenomena which are related to the interaction of a solid body and a fluid are of great interest in recent years. The fluid – structure interaction plays an important role in many engineering applications. One of them is the hydraulic turbine design. The key component of the turbine is the runner, which converts the potential energy of water into mechanical energy. Its failure during the operation time would cause huge damage of the

whole turbine and losses of production which would lead to lost profits. Due to the presence of various excitation forces mainly caused by rotor-stator interaction (RSI), the dynamic response of the runner under operation conditions needs to be investigated. [1] Since operation of the hydraulic turbines in very broad range is required nowadays, turbines experience significant dynamic loading, especially in part load regimes. This loading is not connected only with RSI, but also with periodic shedding due to boundary layer separation, i. e. Kármán vortex street, fur-

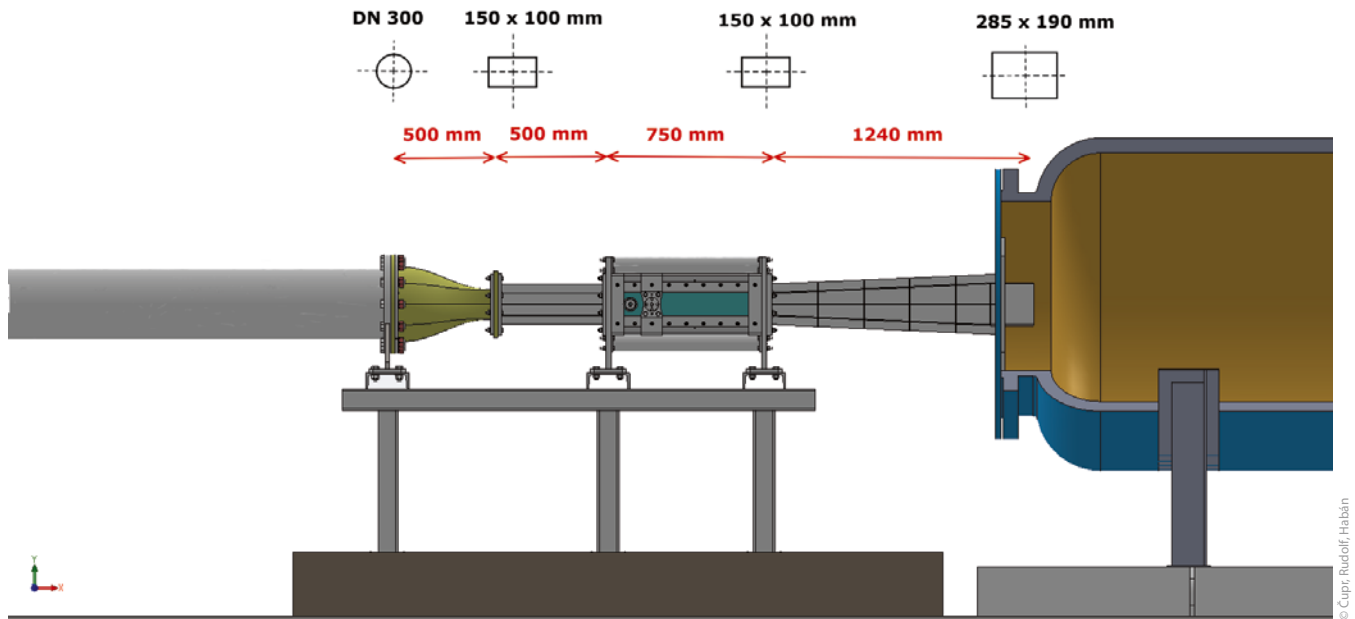


Figure 1: Cavitation tunnel at Brno University of Technology

thermore with massive cavitation in form of cavitation clouds shedding and cavitating vortices as well as with self-induced instability of swirling flow in the draft tube.

To better understand the dynamic response investigation, only one simple blade placed into the cavitation tunnel test section is considered and its dynamic behaviour is studied numerically. The cavitation tunnel was designed and built in our laboratory at Viktor Kaplan Department of Fluid Engineering at Brno University of Technology (**Figure 1**). The test section has a rectangular cross-section with dimensions 100 x 150 mm. Its length is 750 mm [2].

In this paper, the preliminary numerical investigations of added mass and added damping effects acting on the fully fixed hydrofoil are presented. The added mass effect is calculated using acoustic modal analysis of the hydrofoil submerged in still water. The added damping effect is evaluated by two-way coupled fluid-structure interaction analysis.

Synopsis

- Cavitation tunnel to study vibrations of hydrofoil under cavitating conditions was built in collaboration with company Litostroj Engineering in laboratory of V. Kaplan Department of Fluid Engineering, Brno University of Technology.
- Computational modal analysis of the hydrofoil NACA 0003 with blunt edge was performed first in air and then in water to assess the natural frequencies and consequently also the added mass of hydrofoil submerged in water.
- Damping in water environment was computed using two-way FSI analysis for an interval of flow velocities.

2 Added Effects Investigation

When an oscillating structure is submerged into a liquid, its dynamic response is altered by the added effects such as added mass, damping and stiffness. Considering the hydrofoil placed in the cavitation tunnel test section, its motion can be modelled as a single degree of freedom (1-DOF) oscillator. The hydrofoil oscillations can be described by equation of motion (1)

$$m_s \ddot{x} + d_s \dot{x} + k_s x = F(t) \quad (1)$$

where m_s is the mass of the hydrofoil, d_s is the structural damping, k_s is the stiffness of the structure and $F(t)$ is the force acting on the hydrofoil from the fluid. The equation can be rewritten in another form (2).

$$(m_s + m_w) \ddot{x} + (d_s + d_f) \dot{x} + (k_s + k_f) x = 0 \quad (2)$$

The following added effects are present: m_w is the added mass of still water, d_f is damping resulting from momentum exchange between the flow and the structure due to the oscillation and k_f is the fluid-added stiffness.

2.1 Added Mass

The added mass effect controls the frequency of oscillations and causes the shift of natural frequencies compared to the values measured in the air. This effect is investigated for the hydrofoil submerged in still water using acoustic modal analysis. Therefore, the equation of motion (2) can be simplified into (3). As the potential flow modelling approach is used, the added mass due to viscous effects is not taken into account. This approximation is acceptable for hydrofoils and turbine blades, because the viscous effects remain weak. The only fluid effect taken into account is the added mass.

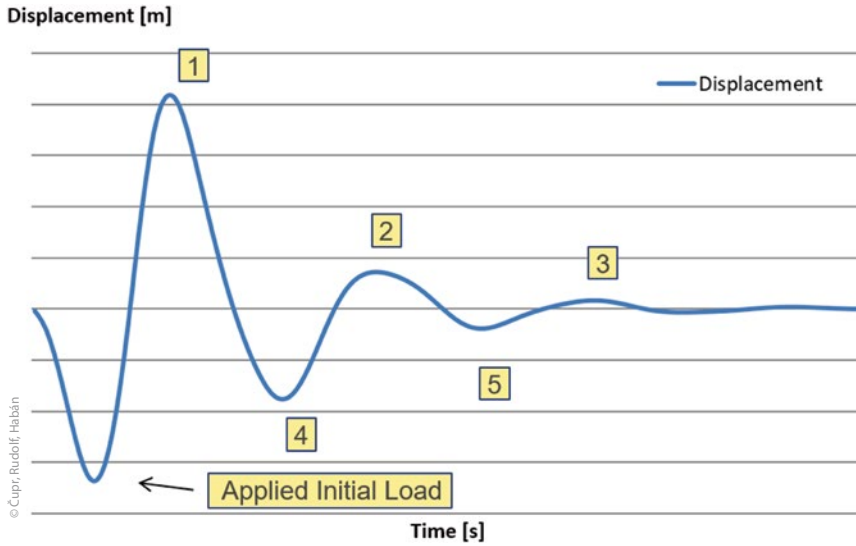


Figure 2: Free oscillations of the hydrofoil

$$(m_s + m_w)\ddot{x} + k_s x = 0 \quad (3)$$

For a lightly damped structure submerged in still water, the first natural frequency can be found as

$$f_{\text{WATER}} = \frac{1}{2\pi} \sqrt{\frac{k_s}{m_s + m_w}} \quad (4)$$

The ratio between the natural frequency in the air f_s and the natural frequency in water f_{WATER} is written as follows

$$\frac{f_{\text{AIR}}}{f_{\text{WATER}}} \cong \frac{\frac{1}{2\pi} \sqrt{\frac{k_s}{m_s}}}{\frac{1}{2\pi} \sqrt{\frac{k_s}{m_s + m_w}}} = \sqrt{\frac{m_s + m_w}{m_s}} \quad (5)$$

Finally, the added mass of the water can be determined as

$$m_w = m_s \cdot \left[\frac{1}{\left(\frac{f_{\text{WATER}}}{f_{\text{AIR}}} \right)^2} - 1 \right] \quad (6)$$

The added mass investigation is carried out using acoustic modal analysis. The first six natural frequencies of the hydrofoil both in the air and in still water are calculated and the added mass of water is calculated using (6) [3],[4].

2.2 Added Damping

If the frequency of excitation forces is close to the natural frequency, the amplitude of displacements rapidly increases. If the excitation frequency is close to the natural frequency and the deformed shape of the structure corresponds to the mode shape, then the resonance occurs. If the resonance is present during the time when the runner is operated, it may lead to fatigue failure or

a crash of the whole runner. According to the magnitude of damping, the amplitude of oscillations can be reduced. The equation of motion for a solid placed in water can be written as follows.

$$(m_s + m_w)\ddot{x} + d_F \dot{x} + (k_s + k_F)x = 0 \quad (7)$$

The structural damping d_s is much lower than fluid damping and can be neglected. The added damping d_F represents the energy extracted from the structure as a result of work done by the fluid flow. The added stiffness describes the change in the flow-induced restoring force with the deflection of the structure.

Assuming, that an oscillating displacement and the velocity displacement are given in (8) and (9), the work $W_{\text{stiffness}}$ performed by the hydrodynamic stiffness force $F_{\text{stiffness}}$ is obtained and given in (10).

$$u = \hat{U} \sin(\omega t - \theta) \quad (8)$$

$$\dot{u} = -\hat{U} \omega \cos(\omega t - \theta) \quad (9)$$

$$W_{\text{stiffness}} = \int_1^2 F_{\text{stiffness}} du = \int_0^T F_{\text{stiffness}} \frac{du}{dt} dt = \quad (10)$$

$$W_{\text{stiffness}} = \int_0^T (k_s + k_{\text{add}}) u \dot{u} dt =$$

$$\hat{U}^2 \omega (k_s + k_{\text{add}}) \int_0^T \sin(\omega t - \theta) \cdot \cos(\omega t - \theta) dt =$$

$$\frac{\hat{U}^2 \omega (k_s + k_{\text{add}})}{2} [\sin^2(\omega t - \theta)]_0^T =$$

$$\frac{\hat{U}^2 \omega (k_s + k_{\text{add}})}{2} (\sin^2(-\theta) - \sin^2(-\theta)) = 0$$

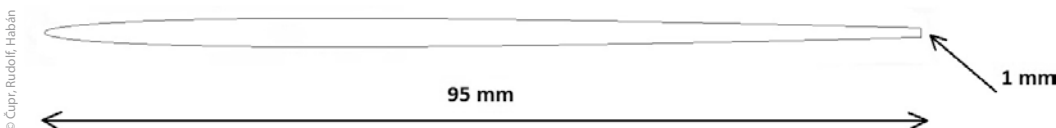


Figure 3: Hydrofoil dimensions

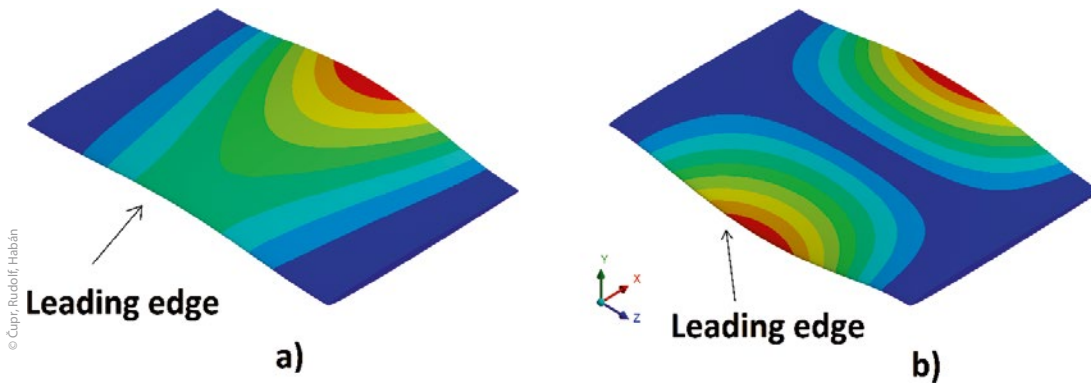


Figure 4: Hydrofoil mode shapes in water a) bending mode (140 Hz) b) torsion mode (305 Hz)

Since the development of work performed by the hydrodynamic inertia force is similar to (10), the only hydrodynamic force performing a work during one oscillating period is the hydrodynamic damping force. Therefore, the investigation of hydrodynamic damping d_F for various flow velocities in the range of 0-30 m/s is presented.

The hydrodynamic damping is investigated using the two-way FSI coupled analysis. At first, a force in the form of pressure load is applied on a top surface of the hydrofoil. Once the initial deflection is reached, the free vibrations are observed (**Figure 2**). The applied pressure is a half sinus wave with frequency corresponding to the first natural frequency of the hydrofoil obtained by acoustic modal analysis. The deflection in the middle point of the trailing edge is monitored. Then the logarithmic decrement Λ and the damping ratio D are calculated using (11) and (12). The final value is calculated as a mean value between the peaks 1-2, 2-3 and 4-5.

$$\Lambda = \frac{1}{n} \ln \frac{x(t)}{x(t + n \cdot T)} \quad (11)$$

$$D = \frac{\Lambda}{\sqrt{(2\pi)^2 + \Lambda^2}} \quad (12)$$

3 Analysis Setup

3.1 Case Description

The hydrofoil used in all numerical cases is the blunt trailing edge NACA 0003 with a chord length of 95 mm and a maximum thickness of 3 mm (**Figure 3**). The trailing edge thickness is 1 mm and the angle of attack is 0°. The hydrofoil is mounted on the shaft which is fitted by bearings and fully fixed on its both ends. The hydrofoil and the shaft are manufactured as a one piece together. For material properties see **Table 1**. The length in span wise direction is 150 mm, which corresponds to the test section depth.

Table 1: Material properties of bronze (Source: [7])

	Density	Young Modulus	Poisson Ratio
Bronze	7 800 kg/m ³	115 GPa	0.33

3.2 Modal Analysis of the Hydrofoil

3.2.1 Modal Analysis in the Air

The first six natural frequencies of the hydrofoil were calculated. All calculations were carried out using commercial software ANSYS Mechanical and CFX 18.1. The structural mesh was created using 17 500 linear hexahedral elements (SOLID 185). The zero displacements in x, y and z directions were prescribed on the hydrofoil surface, where the shaft meets the both hydrofoil sides. The frictionless support, which prescribes the zero displacement in the normal surface direction, was applied on the rest of the hydrofoil sides to keep them planar. This corresponds to real boundary conditions in the cavitation tunnel. For results overview see **Table 2**.

3.2.2 Acoustic Modal Analysis

As the added mass effect influences the dynamic response of the hydrofoil, the natural frequencies of hydrofoil submerged in still water need to be calculated. The acoustic domain corresponds to the dimensions of the test section. The fluid domain is represented by 328 000 hexahedral acoustic elements. Some additional boundary conditions were added to the structural boundary conditions described in the previous case. The FSI Interface was set up on the hydrofoil surface, which was in contact with the water. The zero value of acoustic pressure was applied at the inlet and outlet of the test section. Then the first six natural frequencies were calculated and compared to the previous results (**Table 2**). The ratio between the natural frequency computed in water and in the air is for each mode different. The most significant frequency drop is reached for the first Eigenmode, where the natural frequency drops to 44 % compared to hydrofoil in the air.

The first bending and torsion mode shapes are shown in the figure below (**Figure 4**). The influence of water is stronger for the first bending mode (140 Hz) than for both the first torsion mode (305 Hz) and for the second bending mode (330 Hz). The added mass is computed using equation (6). The hydrofoil mass is 0.264 kg.

3.3 Vortex Shedding Frequency Calculation

When the hydrofoil is placed into the flow, a periodic excitation lift force is acting on its surface due to the vortex shedding behind the trailing edge (von Kármán vortex street). The

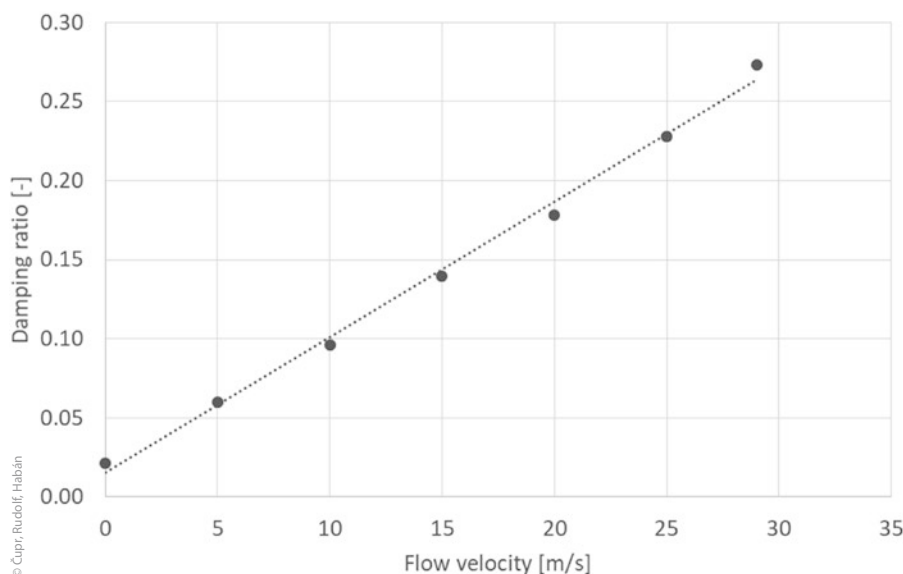


Figure 5: Evolution of hydrodynamic damping ratio according to the reduced velocity

Table 2: Modal analysis results - added mass investigation (Source: Čupr, Rudolf, Habán)

Mode	Natural frequency		Ratio water/air	Added mass	Added mass/ hydrofoil mass
	Air	Water			
	[Hz]	[Hz]	[-]	[kg]	[-]
1	321	140	43.6 %	1.12	4.26
2	525	305	58.1 %	0.52	1.96
3	644	330	51.2 %	0.74	2.82
4	1010	578	57.2 %	0.54	2.05
5	1280	776	60.6 %	0.45	1.72
6	1592	992	62.3 %	0.42	1.58

excitation frequency can be calculated using dimensionless Strouhal number given by

$$St = \frac{f \cdot L}{U} \quad (13)$$

where f is frequency of vortex shedding, L is characteristic length (thickness of the trailing edge) and U is bulk flow velocity. The critical value of the Strouhal number for which the vortex shedding can be observed is around 0.2 [5], [6]. As a preliminary calculation for the coupled approach, the CFD analysis of the flow around the hydrofoil was carried out to determine the vortex shedding frequency for various flow velocities (10 m/s, 20 m/s and 30 m/s). The fluid domain dimensions correspond to the dimensions of the test section, except the downstream length, which is 0.3 m to prevent ill-posed boundary conditions. The domain is meshed by 1 092 000 quadrilateral cells. The analysis was carried out using k- ϵ realizable turbulence model and time step $3.78 \cdot 10^{-6}$ s which provides 44 discrete points per oscillation (flow velocity 30 m/s, vortex shedding frequency about 6000 Hz). At the inlet of the domain, three values of the flow velocity were prescribed. At the outlet of the domain, zero

static gauge pressure was applied. At the sides of the domain, free slip boundary condition was set up. The hydrofoil surface was considered as a wall with no-slip boundary condition. The vortex shedding frequency was calculated using Fast Fourier Transformation (FFT) from the pressure fluctuations monitored behind the trailing edge. Results are summarized in **Table 3**.

3.4 Coupled Analysis

To determine the hydrodynamic damping, the two-way coupled analysis was carried out. This technique was implemented in ANSYS software. The structural and fluid solvers run parallel and exchange data at synchronisation points during the time steps. These values are exchanged on the hydrofoil surface which is in contact with the fluid through the FSI Interface boundary condition. The fluid solver sends the forces which are acting from the fluid on the hydrofoil surface and the structural solver sends back displacements. The half sinus wave pressure load is applied on the hydrofoil top surface to deflect the hydrofoil. Then the free vibrations are observed and hydrodynamic damping from various peaks is evaluated. The structural and fluid meshes are the same as in

Table 3: Vortex shedding frequency (Source: Čupr, Rudolf, Habán)

Flow velocity [m/s]	Vortex shedding frequency [Hz]
10	1611
20	3352
30	5672

previous calculations. The hydrofoil was fully fixed on its both sides. At the inlet of the domain, various flow velocity values were prescribed in range of 0-30 m/s. At the outlet of the domain, zero static gauge pressure was applied. At the sides of the domain, free slip boundary condition was set up. The hydrofoil surface was considered as a wall with no-slip boundary condition and the FSI Interface was applied on its surface. The coupled analysis was carried out using k- ϵ realizable turbulence model within CFX solver. The number of time steps per hydrofoil period is 70 which corresponds to the time step size $\Delta t = 6.75 \cdot 10^{-5}$ s. The peaks of the free vibrations were detected as shown in Figure 2 and the hydrodynamic damping was evaluated using equations (11) and (12). The results are plotted in the following figure (Figure 5). The linear dependence of hydrodynamic damping ratio according to the flow velocity was found.

4 Conclusion

The dynamic response of oscillating hydrofoil placed into the cavitation tunnel test section was investigated. The preliminary CFD analysis was carried out to determine the vortex shedding frequency. Afterwards the modal analysis of the hydrofoil both in the air and submerged in water was performed. The first six natural frequencies and added mass of water were evaluated. The preliminary calculation of hydrodynamic damping using coupled analysis was also carried out. Methodology is based on the initial deflection of the hydrofoil by pressure load applied on the top surface of the hydrofoil and then releasing it and observing free vibrations. Finally, the logarithmic decrement and hydrodynamic damping ratio were calculated. Results show that natural frequency of the dominant bending mode in water is well below the vortex shedding frequencies for most of the operating range. However, this fact might change if the effect of cavitation will be included and when periodic loading by the separated boundary layer will be taken into account for different angles of attack.

Pavel Čupr, Pavel Rudolf und Vladimír Habán

Numerische Untersuchung des „Added-Mass-Effects“ auf einen Tragflügel in einem Kavitationstunnel

Die Effekte „zusätzlicher Masse“ (added mass effect) und „zusätzliche Dämpfung“ (added damping effect) haben großen Einfluss auf das dynamische Verhalten einer von Wasser umgebenen Struktur, wie zum Beispiel die Bauteile einer Turbine. Das Hauptziel der gegenständlichen Arbeit war, vorläufige Ergebnisse numerischer Untersuchungen hinsichtlich dieser beiden Effekte bei Umströmung eines Einzeltragflügels in Wasser darzulegen. Die Berechnung des Effekts zufolge zusätzlicher Masse wurde durch akustische Modalanalyse des Tragflügels in Luft und Wasser durchgeführt. Die Ermittlung der Auswirkungen des Effekts zusätzlicher Dämpfung erfolgte durch eine Zwei-Wege-Fluid-Struktur-Interaktion.

Acknowledgement

This work has been supported by the EU project Computer Simulations for Effective Low-Emission Energy Engineering funded as project No. CZ.02.1.01/0.0/0.0/16_026/0008392 by Czech Republic Operational Programme: Research, Development and Education, Priority axis 1: Strengthening capacities for high-quality research and the collaboration and by the grant no. TH02020705 “Research of turbine blade oscillations with respect to extended regulation range for guarantee of stability and safety of energetic system”

Authors

Ing. Pavel Čupr

Associate professor Pavel Rudolf, Ph. D.

Associate professor Vladimír Habán, Ph. D.

Brno University of Technology

Viktor Kaplan Department of Fluid Engineering

Technická 2896/2

616 69 Brno, Czech Republic

cupr@fme.vutbr.cz

rudolf@fme.vutbr.cz

haban@fme.vutbr.cz

Literature

- [1] Nennemann, B.; Monette, CH.; Chamberland-Lauzon, J.: Hydrodynamic Damping and Stiffness Prediction in Francis Turbine Runners Using CFD. In: IOP Conference Series Earth and Environmental Science 49 (2006), Nr. 7.
- [2] Čupr, P.; Cardoso, M.; Rudolf, P.: Design of a New Cavitation Tunnel at Brno University of Technology for FSI Testing. In: Conference proceeding: Engineering Mechanics, 2018.
- [3] Roth, S.: Identification of Hydrodynamic Damping of Runner Blade Vibrations Using Numerical Simulations. Master thesis, 2008.
- [4] De La Torre, O.; Escaler, X.; Egusquiza, E.; Farhat, M.: Experimental Investigation of Added Mass Effects on a Hydrofoil under Cavitation Conditions. In: Journal of Fluids and Structures 39 (2013), p. 173-187.
- [5] Dörfler, P.; Sick, M.; Coutu, A.: Flow-induced pulsation and vibration in hydroelectric machinery: engineer's guidebook for planning, design and troubleshooting. New York: Springer, 2012.
- [6] Locky, K. J. et al.: Flow-induced vibrations at stayvanes: experience on site and CFD simulations. In: International Journal on Hydropower & Dams 13 (2006), p. 102-106.
- [7] MatWeb (ed.): Material Property Data - Bronze (<http://www.matweb.com/search/DataSheet.aspx?MatGUID=66575ff2cd5249c49d76df15b47dbca4>); view 05.08.2019)